sURL-3K Annotation Guidelines

A Manual for Contextualized Scholarly URL Extraction

Supplementary material for the accompanying paper. Double-blind version.

This manual documents how sURL-3K was annotated. sURL-3K is a benchmark of scholarly URLs paired with the context that gives each URL its meaning, built to support downstream analyses such as link rot studies, meta-analyses, and reproducibility work. Because those analyses depend on the role a URL plays in a paper rather than on the URL string alone, every URL in sURL-3K is annotated together with its target sentence, the neighboring sentences, its location in the document, and one class from an open access datasets and software (OADS) scheme. What follows states the annotation task, the rules that decide what is annotated, the record each instance takes, the procedure the annotator follows, the convention for resolving footnote and reference markers, and the OADS scheme. It is written so that the annotation of any paper can be reproduced from these rules and any released instance can be audited against them.

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1 The Annotation Task

The annotator reads a scholarly paper one rendered page at a time and, for every URL that appears in the author content, records the URL, where it sits in the document, the sentence that gives it meaning, the two sentences that surround that one, and an OADS class. When a URL sits in a footnote or a reference, its meaning is carried by a sentence elsewhere in the paper that points to the URL only through a marker. In that case the annotator resolves the marker and restores the footnote or reference entry into the citing sentence, so that the recorded sentence reads as if the URL had been written in place.

The unit of annotation is a single *instance*, one URL bonded to one target sentence. This matters whenever a sentence carries more than one URL, or when one footnote or reference is cited from more than one sentence. In either case the same sentence appears in several instances, each carrying a different URL or a different restored sentence. sURL-3K contains 3,347 such instances drawn from 352 papers across 11 disciplines. Table 1 gives the breakdown by corpus and document location; every count in this manual is taken from the released annotation table.

Table 1: Scale of sURL-3K by corpus and document location.

	arXiv	CORD-19	Total
Papers	192	160	352
Instances	1,306	2,041	3,347
Instances by document location			
Body	278	568	846
Footnote	638	15	653
Reference	390	1,458	1,848

2 Source Material

sURL-3K is built on two scholarly PDF corpora that differ in layout, citation style, and URL placement, so that a single set of rules is exercised against a wide range of conditions. Of the 352 papers, 192 are from arXiv and 160 are from CORD-19. arXiv papers are mostly single-column and span preprints, conference proceedings, and journal articles; CORD-19 papers are mostly two-column journal articles. Between them the two corpora cover numeric and author-year citation styles. URL line wrapping is far more common in CORD-19, while footnote URLs are common in arXiv and rare in CORD-19. Papers were chosen to exercise these conditions rather than sampled at random, so the corpus is denser in difficult cases, such as scheme-less URLs, wrapped URLs, and grouped citation markers, than a random draw would be. An annotator does not reproduce the selection; it is described here only so a reviewer understands the corpus composition.

Throughout, the annotator works from the page as a reader sees it, not from text converted out of the PDF. Reading the rendered page keeps the typographical and layout signals that conversion discards, above all the superscript markers that tie footnotes and references to the body and the column structure that separates one location from another. The annotator transcribes the *text-layer* URL, the URL that appears as visible characters on the page. A URL that exists only as a clickable hyperlink in the annotation layer of the PDF, with no visible character string, is outside the scope of sURL-3K.

3 Document Locations and Annotation Scope

We partition a paper into author content and boilerplate, and annotate only the author content. The author content is divided into three document locations, and every instance carries exactly one of them.

Body

The running text of the paper, taken to include the abstract, every section from the introduction through the Acknowledgements and Appendices, display text such as titles and section headings, figure and table captions, table content, and equations. A body URL appears inside its own target sentence.

Footnote
The notes set at the foot of a page. A footnote URL usually sits on the same page as the sentence that cites it but apart from it, tied to the body by a superscript or symbolic marker.

Reference
The entries of the bibliography at the end of the paper. A reference URL usually sits on a different page from the sentence that cites it, tied to the body by a numeric or author-year marker.

Everything outside these three locations is publisher boilerplate, which carries metadata such as license statements and ORCID identifiers together with links that do not belong to the paper's argument, such as a link to a citation-validation service or a citation-string generator. The annotator identifies the location of each URL and keeps only those in the author content. Figure 1 shows the distinction on the front page of one paper.

Figure 1: Front page of a representative paper (PMC4567293, CORD-19). In-scope author-content URLs are outlined in **green**, excluded boilerplate URLs in **red**. The publisher header, license block, and citation-service links are boilerplate and are not annotated even though they are valid URLs.

4 What Counts as a URL

Within the author content, four rules decide whether a string is annotated. A URL qualifies when its path is directly represented in the text layer as visible characters, without reconstruction or inference. It is transcribed in whatever surface form it takes, whether it carries a scheme such as `http` or `ftp` or is a scheme-less host or host-path string such as `github.com/user/repo`, and whether or not it carries a port, a non-Latin host, a query introduced by `?`, a fragment introduced by `#`, or a percent-encoded character such as `%20`.

Three kinds of string are set aside. A bare scholarly identifier that is not written as a full URL is excluded, so a DOI of the form `10.NNNN/...` or an arXiv identifier of the form `arXiv:NNNN.NNNNN` is annotated only when it appears as a complete URL. A placeholder such as `http://example.com` is excluded, as is any string that resembles a host but names something rather than locating a resource.

5 The Annotation Record

Each instance is one record with the fields in Table 2. sURL-3K releases the records as a table with one row per instance, held separately from the source PDFs. The JSON below is the same record shown field by field.

Table 2: Fields of one sURL-3K instance.

Field	Meaning
`paper_id`	Source paper identifier (an arXiv id or a PMC id).
`url`	The URL, transcribed verbatim, with any wrap rejoined into one string.
`location`	One of `body`, `footnote`, `reference`.
`target`	The target sentence. For a body URL, the sentence carrying the URL. For a footnote or reference URL, the citing sentence once the marker is resolved and the entry restored.
`preceding`	The sentence before the target in the same paragraph, empty at a paragraph boundary.
`trailing`	The sentence after the target in the same paragraph, empty at a paragraph boundary.
`oads_class`	One of the six OADS classes of Section 7.

The record below is a body instance from paper `1202.3074`, with every value verbatim from sURL-3K.


```
{
  "paper_id": "1202.3074",
  "url": "http://www.cs.wisc.edu/condor",
  "location": "body",
  "preceding": "It provides support for creating Condor dagman job files, thereby converting
      the iterations of loops into different jobs.",
  "target": "Condor (http://www.cs.wisc.edu/condor) is a job management system developed at
      the Computer Science Department of the University of Wisconsin.",
  "trailing": "Python is a free, open source, cross-platform programming language and has
      become a viable alternative to Matlab and Octave recently.",
  "oads_class": "third-party-software"
}
```


The target sentence always has a value. The preceding and trailing sentences come from the same paragraph as the target, and either is left empty when the target begins or ends its paragraph. In the record above both neighbors are present; when the target is the last sentence of its paragraph the trailing field is empty, and likewise for the preceding field at the start of a paragraph.

6 Annotation Procedure

For each paper the annotator moves page by page. On every page the annotator finds the visible URLs in the author content and, for each one, transcribes the URL verbatim, decides its document location, captures its context in the way that location requires, and assigns one OADS class using the scheme of Section 7. When a URL is split across two lines, two columns, or two pages, the parts are rejoined into one string, taking care to add no character that the wrap introduced. Every hyphen is kept as it appears, and a hyphen is dropped only when the line wrap itself inserted it and the URL continues as the same uninterrupted token. A target sentence and its neighbors may themselves cross a column gutter or a page boundary, and the annotator reads across the boundary to capture the whole sentence. How the context is captured, and how one sentence can give several instances, differs by location, as the three parts below describe.

For a footnote or reference URL, whose context is not beside the URL, the annotator restores it. Restoration puts the full footnote or reference entry that carries the URL where its marker stood, so that the whole entry, and with it the URL, joins the citing sentence and forms the target sentence. The entry is restored in full and is not reduced to its URL, and its own leading label, such as a footnote number or a bracketed reference number, is dropped. Nothing else in the sentence changes, and the annotator sets the restored entry inside the punctuation the sentence already uses so the result reads as a single sentence. When a marker groups or ranges over several keys, only the key that points to the URL-bearing entry is replaced, the sibling keys stay as they are, and each URL-bearing key is handled on its own.

6.1 Body

A body URL sits inside its own target sentence. The annotator records that sentence together with the preceding and trailing sentences of the same paragraph, leaving a neighbor empty when the target begins or ends its paragraph. When one body sentence carries more than one URL, each URL gives its own instance sharing that sentence. In the example of Table 3, the URL is the archaeopteryx tree viewer, a tool made by others, so its class is `third-party-software`.

Table 3: A body URL and its context, paper 1107.5095.

Target sentence	Graphics made with ggplot2 [18] and the archaeopteryx tree viewer (**http://www.phylosoft.org/archaeopteryx/**).
Preceding sentence	Aaron Gallagher co-developed the guppy framework and helped write the simulation code to validate squash clustering.
Trailing sentence	The first author was supported in part by NIH grant HG005966-01, and the second author was supported in part by NSF grant DMS-09-07630.

6.2 Footnote

A footnote URL sits at the foot of the page, apart from the sentence that cites it and tied to it by a superscript or symbolic marker, in one of the forms below. The annotator detects the marker, finds the footnote it points to, and restores the full footnote into the citing sentence to form the target sentence, then records the preceding and trailing sentences of that citing sentence. When one sentence carries several URL-bearing footnotes, each footnote gives its own instance sharing that sentence.

Footnote marker form	**Example**
Superscript numeral	a word with a raised `1`
Symbolic mark	`*`, †, ‡

The instance in Table 4, from paper `1204.0171`, has a superscript marker on *Corel Dataset* that ties the sentence to a footnote stating where the dataset is available. The whole footnote, its descriptive text and its URL together, replaces the marker. The class is `third-party-dataset`, since the Corel Dataset is a standard benchmark created by others and the authors only re-host a copy, and hosting does not change provenance.

Table 4: Footnote restoration, paper 1204.0171.

Citing sentence	In this section, the algorithms have been analyzed on two image classification benchmark datasets, which are Corel Dataset[2] consisting of 599 classes and Caltech 101 Dataset consisting of 102 classes.
Footnote	[2] The dataset is available on `https://github.com/meteozay/Corel_Dataset.git`
Restored target sentence	In this section, the algorithms have been analyzed on two image classification benchmark datasets, which are Corel Dataset (**The dataset is available on** `https://github.com/meteozay/Corel_Dataset.git`) consisting of 599 classes and Caltech 101 Dataset consisting of 102 classes.

A single sentence may carry more than one footnote marker, and each URL-bearing footnote then gives its own instance. The sentence in Table 5, from paper `1011.3595`, carries three superscript markers, one on each of three named resources, so it recurs as three instances that differ only in which footnote is restored. All three URLs are `general-url`.

Table 5: Footnote restoration for a sentence with three markers, paper 1011.3595.

Citing sentence	For example, the Triplify update vocabulary[16] alerts data consumers to incremental changes by providing pointers to RDF documents which have changed, while DSNotify[17] and the Web of Data Link Maintenance Protocol[18] facilitate synchronization of Linked Data link sets.
Footnotes	[16] `http://triplify.org/vocabulary/update` [17] `http://dsnotify.org/` [18] `http://www4.wiwiss.fu-berlin.de/bizer/silk/wodlmp/`
Restored target sentence	1. For example, the Triplify update vocabulary (**http://triplify.org/vocabulary/update**) alerts data consumers to incremental changes by providing pointers to RDF documents which have changed, while DSNotify and the Web of Data Link Maintenance Protocol facilitate synchronization of Linked Data link sets. 2. For example, the Triplify update vocabulary alerts data consumers to incremental changes by providing pointers to RDF documents which have changed, while DSNotify (**http://dsnotify.org/**) and the Web of Data Link Maintenance Protocol facilitate synchronization of Linked Data link sets. 3. For example, the Triplify update vocabulary alerts data consumers to incremental changes by providing pointers to RDF documents which have changed, while DSNotify and the Web of Data Link Maintenance Protocol (**http://www4.wiwiss.fu-berlin.de/bizer/silk/wodlmp/**) facilitate synchronization of Linked Data link sets.

6.3 Reference

A reference URL sits in the bibliography, apart from the sentence that cites it and tied to it by a numeric or author-year marker, in one of the forms below. The annotator detects the marker, finds the reference entry it points to, and restores the full entry into the citing sentence to form the target sentence, then records the neighbors. A grouped or ranged marker expands to its individual references, and each URL-bearing entry gives its own instance, while a marker whose entry carries no URL gives none. A reference cited from several sentences gives one instance per citing sentence.

Reference marker form	**Example**
Bracketed numeral	`[4]` or `4`
Grouped numerals	`[1, 15, 4]`
Ranged numerals	`[21-24]`
Author-year	`(Arnett, 2000)`, `Arnett (2000)`

The example in Table 6, from paper `1005.4552`, has a sentence with a grouped numeric marker and a second numeric marker, where three of the cited references carry a URL. It shows grouped-marker expansion, one instance per URL-bearing key, and a key whose entry has no URL (reference 4) giving no instance. Reference 1 is `general-url`, and references 15 and 21 are `third-party-software`.

A reference marker can also take a ranged form such as `[21-24]`, which stands for every reference in the range. The example in Table 7, from paper `1105.1815`, cites references 21 to 23 with

Table 6: Reference restoration with a grouped marker, paper 1005.4552.

Citing sentence	Indeed, there are already various online repositories of formal proofs [1, 15, 4] and catalogs of deductive tools [21].

Reference entries with a URL	`[1]` The Archive of Formal Proofs. `http://afp.sourceforge.net/`
	`[15]` The Mizar Mathematical Library. `http://mizar.org/library/`
	`[21]` The QPQ Deductive Software Repository. `http://www.qpq.org`
	(reference 4 carries no URL)

Restored target sentence	1. Indeed, there are already various online repositories of formal proofs [(**The Archive of Formal Proofs, http://afp.sourceforge.net/**), 15, 4] and catalogs of deductive tools [21]. 2. Indeed, there are already various online repositories of formal proofs [1, (**The Mizar Mathematical Library, http://mizar.org/library/**), 4] and catalogs of deductive tools [21]. 3. Indeed, there are already various online repositories of formal proofs [1, 15, 4] and catalogs of deductive tools [(**The QPQ Deductive Software Repository, http://www.qpq.org**)].

the single marker `[21-23]`, and all three carry a URL. The range first expands to its individual references, and each URL-bearing one is then restored on its own. References 21 and 23 are `general-url`, and reference 22 is `third-party-dataset`.

Table 7: Reference restoration with a ranged marker, paper 1105.1815.

Citing sentence	Sources: [21–23].
Range expansion	The range `[21-23]` first expands to the individual references `[21, 22, 23]`, so each one is looked up and restored on its own.

Reference entries with a URL	`[21]` A list of historical magnetic hard drive sizes and prices can be found at `http://ns1758.ca/winch/winchest.html`
	`[22]` More recent data on historical magnetic hard drive sizes and prices can be found at `http://www.mattscomputertrends.com/harddiskdata.html`
	`[23]` A detailed chronological history of solid state drive development can be found at `http://www.storagesearch.com/`

Restored target sentence	1. Sources: [(**A list of historical magnetic hard drive sizes and prices can be found at http://ns1758.ca/winch/winchest.html**), 22, 23]. 2. Sources: [21, (**More recent data on historical magnetic hard drive sizes and prices can be found at http://www.mattscomputertrends.com/harddiskdata.html**), 23]. 3. Sources: [21, 22, (**A detailed chronological history of solid state drive development can be found at http://www.storagesearch.com/**)].

7 OADS Class Scheme

Each URL is labeled with one of six classes from an established OADS scheme. This section gives the generalized scheme the annotator applied, stated at the level of the signals that decide a class rather than as an enumeration of hosts, so that it holds across the heterogeneous sources sURL-3K draws from.

Four of the six classes follow from two independent dimensions. The first is the resource type, either a dataset or a piece of software. The second is the provenance, either the paper's own authors or a third party. Crossing them gives a third-party dataset, an author-provided dataset, a third-party software, and an author-provided software. A fifth class, `project`, covers a URL that bundles more than one artifact type, and the sixth, `general-url`, covers everything that is neither a dataset, software, nor a project. The first five are the OADS classes; `general-url` is the non-OADS class. Table 8 states each class with one real example.

The annotator settles a class from the context first and from the URL string only as a fallback. The most reliable signal is the verb around the URL, not the noun. A data verb such as *downloaded from*, *obtained from*, *retrieved from*, *queried*, or *deposited in* points to a dataset, while a software verb such as *implemented in*, *used to*, *ran with*, *computed using*, or *invoked* points to software. A first-person voice paired with a sharing verb, such as *we release*, *we make available*, or *our implementation*, marks the resource as author-provided. When the same URL is described as holding both code and data, or is named as a project, the class is `project`.

Some nouns do not settle the type on their own. Words such as *framework*, *pipeline*, *model*, *system*, *method*, and *approach* may name a runnable artifact or an idea presented in the paper. When the URL is a paper, preprint, or DOI used in a conceptual sense, the class is `general-url`; when the URL reaches a runnable artifact described as usable, with versions, install or run instructions, or a package index, the class is software. Two rules stand above the rest. Hosting is not provenance, so a repository deposited by the authors is author-provided even on a third-

Table 8: The six OADS classes, each with a generalized definition and one canonical example target sentence from the OADS labeling scheme.

Class	Definition	Example target sentence
`third-party-dataset`	A collection of data (dataset, corpus, benchmark, archive, or database) created by someone other than the authors, which they use, download, retrieve, query, or consult.	The NPI dataset was downloaded from `https://openpsychometrics.org/`.
`author-provided-dataset`	A collection of data created or generated by the authors and shared as part of the study. Hosting in a third-party repository does not change its provenance.	To this end, we have made the entire dataset used in this paper publicly available on the web at `http://neuro.gatech.edu/~potter/public-data`.
`third-party-software`	A computational or executable artifact (tool, library, package, framework, service, API, or model) created by others and invoked, installed, or used by the authors.	Venn diagrams were created using VENNY 2.0.2 (`http://bioinfogp.cnb.csic.es/tools/venny/index.html`).
`author-provided-software`	An artifact of the same kinds, created by the authors and released as part of the study.	The code is openly available at `https://github.com/alexis-roche/variational_sampler`.
`project`	One URL that gathers more than one artifact type, showing both data and software cues, or framed in the text as a research project tied to such artifacts.	The source code, data, and documentation are available on GitHub at `https://github.com/LabAdvComp/matsu-project`.
`general-url`	Anything that is neither a dataset, software, nor a project, such as a paper or DOI landing page, a preprint, a homepage, an informational page, a policy, or a blog post.	These panels, produced by Quality Control for Molecular Diagnostics (`http://www.qcmd.org`), enable laboratories to process and test proficiency samples using the same protocol they use for clinical specimens.

party host, and a repository owned by an unrelated group is third-party even when its contents are data or software; provenance is read from the voice and the verb, never from the host. And a URL embedded in a bibliographic citation, surrounded by author names, a title, a venue, a year, or page numbers, points to the paper and is labeled `general-url`, even when the cited paper is itself about a dataset or a piece of software. When two classes remain equally plausible, the annotator takes the more likely one rather than a fixed default.